

scripts/autonomous-qa/record-qa-session.sh — User Guide

Last verified: 2026-07-26 (LVA-018 script-docs backfill cycle)
Inheritance: HelixConstitution §11.4.18 (script docs); Lava §6.J / §6.Z (anti-bluff evidence) **Classification:** project-specific

Overview

Records the Android emulator screen during an autonomous QA run and saves a **verified** MP4 to ~/Downloads/lava-qa-<timestamp>-<scenario>.mp4.

The recorder uses adb shell screenrecord in the background. Because screenrecord is capped at 180 seconds per invocation, long tests are recorded as sequential chunks that are concatenated with ffmpeg after the test command exits. Chunks are staged under /tmp so large captures never bloat the repo.

Anti-bluff contract (§6.J / §6.Z)

The script **refuses to claim success** unless the recorded file is non-empty AND ffmpeg confirms it contains a video stream with at least one decoded frame. An empty/corrupt/unprobed recording is exit 2, never a silent "recorded".

Usage

```
scripts/autonomous-qa/record-qa-session.sh --serial <serial> --
scenario <name> [options]
```

Flag	Required	Default	Meaning
--serial <serial>	yes	—	adb serial of the running emulator/device
--scenario <name>	yes	—	short scenario slug for the filename (normalized to lowercase alphanumerics + hyphen)
	no	sleep 5	

Flag	Required	Default	Meaning
--test-command <cmd>			shell command to run while recording
--output-dir <dir>	no	~/Downloads	directory for the final MP4
--bit-rate <bps>	no	4000000	screenrecord bit-rate
--chunk-sec <secs>	no	180	chunk length; MUST be $\leq$ 180 (screenrecord limit)
-h --help	no	—	show help

Environment: ANDROID_HOME (adb location), FFMPEG (ffmpeg path override).

Exit codes

- 0 — test command passed AND video verified
- 1 — test command failed BUT video was verified (the recording is still valid evidence of the failure)
- 2 — setup error (missing args, adb/ffmpeg unavailable, device unreachable) or video verification failed

Examples

```
# Record a Challenge run end-to-end
scripts/autonomous-qa/record-qa-session.sh \
  --serial 127.0.0.1:42317 --scenario search-flow \
  --test-command './gradlew :app:connectedDebugAndroidTest'
  -Pandroid.testInstrumentationRunnerArguments.class=java.app.challenges.Challenge

# Quick 5-second smoke recording
scripts/autonomous-qa/record-qa-session.sh --serial emulator-5554
  --scenario smoke
```

Companion files

- scripts/autonomous-qa/run-iteration.sh — embeds the same chunked-recording pattern inline for matrix iterations
- scripts/autonomous-qa/vision-analyze.sh — analyzes the recordings this produces
- scripts/record-challenge-video.sh / scripts/record-device-session.sh (+ their docs) — the sibling recording paths